

1. The solution should allow configuration of different optimization scenarios for the production process (manufacturing/packaging) using a variety of criteria.
2. The solution will provide the ability to adjust the production line schedule based on real-time monitoring of production line performance.
3. The solution will facilitate decision-making for fast adjustments of the production/packaging schedule in case of delays or equipment downtime. For example, when production “freezes” for 10 hours, should the remaining part of the order be moved to the next cycle or not?
4. The solution will incorporate DEMO’s order-ranking algorithm and allow adjustment of different parameters (e.g., weights).
5. The solution will take into account changeover times, the maintenance plan, the machine’s characteristics/operating status, and campaign/format-change constraints.
6. The solution will leverage available data from Photocells monitoring platform in order to monitor the number of products/production speed, and data from ProdAction in order to determine the number of products per batch.
7. The solution will run a regular process to identify any deviations in the production schedule compared to actual execution, with the purpose of informing the user via notifications and proposing alternative schedules.
8. The solution should follow a scheduling logic that groups orders, prioritizes them, and “cuts” them into campaigns, respecting the format-change rule (during a production or packaging cycle, each format is applied only once in the Work Center—or not at all if there are no orders for the period).
9. The solution should provide fully configurable changeover times between different product codes/machines, selection of backup machines in case of malfunctions, and the ability to add/modify production lines and product codes.
10. The algorithm is designed to meet one or more business-critical objectives, such as maximizing demand fulfillment and minimizing changeover time and idle time.
11. The solution will calculate:

Production Schedule: Detailed allocation of orders by machine and time period, including the product, format, and volume.

Packaging Schedule: Optimized packaging plan by machine and time period, aligned with production (or finished-goods inventory), the start of the packaging order time window (pack slot), and packaging codes per customer/country.

12. The calculated schedule will be able to be sent to SAP.

13. For Line Supervisors, the user interface will focus on presenting the production schedule based on shifts per machine and workstation, helping supervisors coordinate tasks effectively (e.g., ordering/issuing raw materials).

14. For the Sales Team, the user interface will focus on a dedicated view that tracks the status of each order throughout the production and packaging lifecycle (including planning and execution).

15. For the Production Team, the interface will offer an interactive “what-if” simulation environment, allowing them to test different production scenarios by adjusting parameters such as machine availability, order priorities, and shift configurations. The dashboard will also provide real-time visibility of actual vs. planned production, enabling dynamic rescheduling when delays or underproduction occur, as well as operational alerts.

16. Using data from Photocells monitoring platform, if some time passes without new packs being detected, an alert will be generated that the process has slowed down (or stopped). Based on the packs produced so far, decide whether to stop this production process (and switch to another order) or continue in the “best possible way.”

17. The solution should support a forecasting horizon of up to 52 weeks.

18. Support packaging and production cycles of different durations.

19. The solution should calculate whether there is available capacity to execute the orders.

20. System integration: Avoid double entry of information that is entered into the system manually.

21. System integration: Timely synchronization of information.

22. Visualization should also be available at the individual line level, beyond the combined visualization of all lines that was demonstrated; statistics should be calculated both per line and per machine.

23. Within the scope of the project, calculations will be performed based on SAP Planned Orders and the appropriate constraints (e.g., BoM availability).

24. **SAP Process Orders** (the schedule to be executed) will be taken into account by the system in schedule re-calculation scenarios.

25. Each line has different resources, often multiple (e.g., two labeling machines). It was also clarified that there are cases where different machines can produce the same product. The system must be able to select the most suitable one based on predefined rules (e.g., based on volume).
26. The bill of material for each finished product will be defined as master data, and based on material availability data it will be determined whether a schedule is feasible.
27. The material-availability check logic is that at the long-term (annual) level orders will be ranked, and then at the quarterly level the feasibility of the schedule should be checked (based on the output from the annual plan) and that schedule should be confirmed/rejected.
28. Production is performed in stages (and machines, e.g., tank, filling machine, etc.), so a failure in one stage negatively affects the progress of production until it is restored. This also affects the schedule for upcoming orders that have already been planned, and it must be taken into account during scheduling.
29. If there is a breakdown that has not been captured in the system, there must be the ability to declare it manually through the platform.
30. There should be the ability to manually edit the schedule before it is sent to SAP.
31. For packaging in glass containers, there is a rule (based on order volume) for which of the four packaging machines should be used, and this was used in Excel.
32. For packaging in glass containers, the rule is that an order should not move to a smaller machine when the packaging machine is overcapacity; instead, a third line (which is a backup) should be opened.
33. For packaging scheduling, the format change is considered only when the container changes, not the packaging. I.e., switching from 20 to 10 and back to 20 vials for packaging is allowed. In practice, this change has a time cost.
34. The PoC analysis should be based on 2025 data (forecast from December 2024).
35. Packaging: Either Farcon or Dividella is in operation, as they are operated by the same team.
36. Packaging: There must be no overlap in format changeovers between machines because they are carried out by the same staff.
37. Packaging: Piramal must receive 1,800,000–2,000,000 amp per month.
38. Packaging: Minor format changes are allowed.

39. For organizations, the order cannot be split (Unicef/Imres/ICRC/IDA).
40. When there is a tie, it is better to distribute production so that one customer is not fulfilled 100% while another is 0%.
41. Packaging: The schedule should start with the week that has the highest total score and then continue with the week with the second-highest score, and so on.
42. Production: One Active Pharmaceutical Ingredient (API) per week.
43. Production: If production of an API exceeds one week, also schedule a second week (not necessarily consecutive).
44. Production: If production of an API does not fill a full week, schedule a second API as well.
45. Production: Consolidation of orders into full campaigns with coverage of 6 months of demand where shelf life is 24 months, and 9 months of demand where shelf life is 36 months.
46. Labeling is scheduled at least 3 days after production, while packaging at least 4 days after production.
47. Take scrap/yield loss per semi-finished product into account when calculating the planned production quantity.
48. Alerts for semi-finished product that goes to packaging after the permitted storage time has elapsed based on shelf life.
49. For semi-finished product T (e.g., EVATON), planning is done for 2 batches at a time (e.g., 2×180 amps) — which lasts less than a campaign. After that, the earliest it can be planned again is after 3 weeks. If a mold change has occurred during those weeks, it must be scheduled in the next cycle.
50. It was requested that when the duration of a batch is shorter than the shift time, the machine should be cleaned immediately and the remaining time should be productive. Example: First shift produces a semi-finished product with a 2-hour production duration. Cleaning takes place as soon as production ends, during the remainder of the shift (8 hours), leaving a second 6-hour shift for production.